Bridging the Generalization Gap in Low-Resource Pakistan Sign Language Recognition using Pretrained Visual–Temporal Models

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Abstract

Sign languages are spatially encoded, inherently dynamic visual languages that serve as the primary communication medium for people with hearing disabilities. Automated Sign Language Recognition Systems (SLRs), which is the translation of gestures into spoken or written text using technology, can significantly reduce the communication barrier between deaf and hearing communities, yet Pakistan Sign Language (PSL), which serves as the medium of communication for more than ten million hearing-impaired citizens in Pakistan, remains severely underrepresented in SLR research. Existing PSL work is largely confined to static alphabet recognition or to dynamic models evaluated without accounting for signer diversity, rendering reported accuracies misleading for real-world deployment. The official PSL Dictionary compounds this by providing only two videos per sign from a single signer, making generalization across signers impossible to train or assess. We address both the data and modelling problems together. We first present **PSL-104**, a signer-diverse dataset of 104 commonly used Urdu words and alphabets (70 dynamic and 34 static gestures), recorded by six signers across varied environments to yield 1,872 original clips, nearly ten times the PSL Dictionary's coverage, and expand it via a spatial and temporal augmentation pipeline to 4,680 training, 1,248 validation, and 1,046 test clips. We then introduce a strictly signer-disjoint evaluation protocol, demonstrating that the random-split methodology prevalent in prior PSL work induces signer leakage that artificially inflates accuracy, a flaw shared by both static alphabet classifiers and dynamic video-based approaches in the existing literature. Under this protocol, we show that models trained from scratch fail to generalize across unseen signers, as evidenced by our own custom 3D-CNN based architecture collapsing, confirming that without a strong pretrained visual prior, scratch-trained models cannot bridge the signer gap regardless of architectural complexity. We then adapt SignVLM [15], a frozen CLIP ViT-L/14 encoder paired with a trainable Efficient Video Learning (EVL) temporal decoder, achieving **83.49% validation** and **83.12% test accuracy** under signer-disjoint conditions. N-shot experiments confirm sample efficiency as 4-shot training alone yields 81.25% validation accuracy, establishing SignVLM as a practical solution for low-resource PSL recognition.

Keywords: Pakistan Sign Language, Sign Language Recognition, Transfer Learning, CLIP, SignVLM, Signer Independence, Low-Resource, Visual-Temporal Transformer, Dataset Construction, Video Augmentation

Contents

1 Introduction

Sign languages are the primary means of communication for people with hearing impairments. An estimated 70 million deaf and/or hard-of-hearing individuals worldwide rely on more than 300 different sign languages to communicate and participate in their communities [1]. Unlike spoken languages, sign languages convey meaning through the coordinated use of handshape, movement, facial expression, and body posture. As a result, they combine both spatial and temporal linguistic information.

Sign Language Recognition (SLR), which aims to map signed video sequences to spoken or written language, has received growing attention because of its potential to reduce communication barriers between Deaf and hearing communities. Applications of SLR include assistive technologies, education, and healthcare. Considerable progress has been achieved for well-resourced sign languages such as American Sign Language (ASL) and Chinese Sign Language (CSL), largely due to the availability of large annotated datasets [23, 31]. In contrast, most of the world's sign languages still lack sufficient labeled data and standardized evaluation practices.

SLR systems are commonly categorized along two dimensions. The first concerns the type of gesture being recognized. *Static* signs consist of a single hand configuration and can often be identified from a still image, whereas *dynamic* signs depend on motion across time and therefore require temporal modelling over video sequences [7]. The second dimension concerns task scope. *Isolated* SLR focuses on recognizing one sign at a time, while *continuous* SLR processes uninterrupted signing streams and must additionally segment them into individual sign units [8]. This work focuses on isolated dynamic SLR, which forms the basis for more advanced continuous recognition systems.

An isolated SLR system is only practically useful if it can recognize signs produced by people who were not seen during training. This property is known as signer independence. Signer-dependent (SD) models, which are trained and evaluated on the same set of signers, often report high accuracy because they learn signer-specific visual cues such as background, skin tone, or hand size. However, these systems frequently fail when applied to new users [10]. For real-world deployment, signer-independent (SI) evaluation, where training and testing signers are completely separated, is therefore essential [9].

Pakistan Sign Language (PSL) is used by an estimated ten million hearing-impaired individuals in Pakistan [1]. Importantly, sign languages are independent linguistic systems and are not derived directly from the spoken languages of their regions. PSL has its own vocabulary, grammar, and phonological structure that differ from Urdu, English, ASL, and other sign languages [5]. Consequently, models trained on ASL or CSL cannot be directly applied to PSL, making dedicated PSL research necessary.

Despite this need, PSL remains significantly underrepresented in SLR research. Zahid et al. [5] presented the first systematic review of machine learning approaches for Urdu and PSL sign language recognition. Their study showed that most previous work focused on static hand configurations representing Urdu alphabet letters, typically using small controlled datasets and signer-dependent evaluation settings. More recent studies have explored dynamic video-based PSL recognition [13, 14]. However, as we demonstrate later, their reported results rely on signer-inclusive evaluation protocols that do not reflect realistic deployment conditions.

A major limitation in PSL research is the lack of suitable data. The official PSL Dictionary [16], currently the only publicly available PSL video resource, contains only two video samples per sign, both recorded by a single signer in controlled studio conditions. Existing PSL datasets also remain limited in vocabulary size and signer diversity [3, 4]. As a result, there is currently no dataset capable of supporting robust signer-independent learning and evaluation.

To address both problems, this paper makes two core contributions. First, we present **PSL-104**, a purpose-built multi-signer PSL video dataset and augmentation pipeline described in detail in Section 3. Second, we adapt and evaluate **SignVLM** [15] on PSL under a strictly signer-disjoint protocol, with full experimental results reported in Section 5.

SignVLM builds on two foundational developments in large-scale visual representation learning. The first is CLIP (Contrastive Language-Image Pretraining) [18], proposed by Radford et al. at OpenAI, which trained a Vision Transformer on 400 million internet image-text pairs using a contrastive objective that aligns visual and linguistic representations in a shared embedding space. The resulting encoder develops remarkably fine-grained visual understanding — including of human hands, body posture, and spatial configurations — without any task-specific supervision, simply because these visual concepts appear abundantly in natural language descriptions of internet imagery. This makes CLIP a uniquely strong foundation for sign language recognition, where the key discriminative signal is subtle variation in hand shape and motion across signers with diverse appearances [**?**, 29]. The second building block is the Efficient Video Learner (EVL) [19], proposed by Lin et al., which demonstrated that a frozen CLIP image encoder can be repurposed for video understanding by adding a lightweight trainable temporal decoder that attends across the sequence of per-frame CLIP embeddings. Because only the temporal decoder is trained from scratch, the approach is highly parameter-efficient and requires far less data than training a full video model end-to-end.

SignVLM [15], proposed by Luqman, applies this CLIP+EVL framework specifically to sign language recognition and was the first work to systematically validate it across multiple sign language benchmarks. Evaluated on KArSL (Arabic SL), WLASL (ASL), LSA64 (Argentinian SL), and AUTSL (Turkish SL), SignVLM outperformed prior state-of-the-art models on three of the four benchmarks, with the performance advantage over purely scratch-trained video models being most pronounced in low-data settings — precisely the regime that characterizes all low-resource sign languages including PSL. The architecture is particularly well-suited to signer-independent recognition because CLIP's pretraining objective — aligning images with natural language descriptions — naturally produces representations that abstract away identity-specific appearance, encoding "a hand performing a thumbs-up" rather than "this particular person's hand performing a thumbs-up." This inherent signer-agnosticism is the property most needed to bridge the generalization gap we document on PSL-104. The full experimental analysis of how and why SignVLM succeeds where scratch-trained architectures fail is presented in

Sections 5 and 6.

Contributions

This paper makes the following contributions:

‘‘‘

1. **PSL-104 dataset:** We introduce a 104-class multi-signer PSL video dataset containing 18 original clips per class (6 signers × 3 videos), recorded at a resolution of 1278×780 with a standardized two-metre recording distance across varied environments and lighting conditions. The dataset contains 1,872 original clips and is the first PSL dataset specifically designed for signer-disjoint training and evaluation.
2. **Video augmentation pipeline:** We develop a spatial, photometric, and temporal augmentation pipeline that includes lower-body cropping, scaling, rotation, brightness adjustment, BSH enhancement, frame jitter, and frame dropping. The resulting augmented dataset contains 4,680 training clips, 1,248 validation clips, and 1,046 test clips.
3. **Signer-disjoint evaluation protocol:** We introduce and strictly enforce a signer-disjoint evaluation protocol for PSL recognition. Our experiments show that the random-split methodology used in previous PSL studies causes signer leakage and can greatly inflate the reported accuracy for models trained from scratch.
4. **SignVLM adaptation for PSL:** We adapt SignVLM to PSL-104 and achieve 83.49% validation accuracy and 83.12% test accuracy under signer-disjoint evaluation, representing the highest reported performance on a genuinely signer-independent PSL benchmark.
5. **N-shot analysis:** We evaluate SignVLM under limited-data settings and show that 4-shot training achieves 81.25% validation accuracy, 8-shot training achieves 84.29%, and full training with 15 original clips per class achieves 85.14%. These results demonstrate the strong sample efficiency of pretrained visual representations.
6. **Future roadmap:** We propose a practical path toward sentence-level PSL translation by combining SignVLM-based word recognition with NLP-based sign-to-text generation [17]. We also outline a dataset expansion strategy targeting 300–500 sign classes. ‘‘‘

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the construction and augmentation of PSL-104. Section 4 presents the model architectures. Section 5 reports experimental findings. Section 6 provides theoretical analysis. Section 7 compares our approach with previous PSL-specific studies. Section 8 discusses limitations and future directions, and Section 9 concludes the paper.

2 Related Work

Research in automated sign language recognition can be broadly divided into three connected areas: dataset construction, recognition architectures, and evaluation methodology. This section reviews prior work in each area and explains how these developments motivate the present study.

PSL and Urdu Sign Language Recognition.

Zahid et al. [5] conducted the first systematic review of machine learning methods applied to PSL and Urdu Sign Language. Their analysis showed that most existing systems rely on static images of Urdu alphabet letters, small single-signer datasets, and classical machine learning methods or shallow CNN architectures. Imran et al. [3] introduced a dataset of PSL hand configurations for Urdu alphabets, consisting of 40 images per class captured from multiple hand orientations, along with a supporting mobile recognition application. Arooj et al. [4] later extended this work using a hybrid CNN-SIFT framework with Kinect sensor data, again focusing on static Urdu alphabets.

Although these studies represent important early efforts, they remain limited to static image classification and cannot model the temporal structure of dynamic PSL gestures, which form the majority of natural PSL vocabulary. Mirza et al. [6] explored video-based PSL recognition using bag-of-words representations combined with SVM classifiers, providing one of the few attempts to move beyond static recognition, though still within a signer-dependent setting.

For dynamic PSL recognition, Hamza and Wali [13] applied video recognition architectures such as C3D, I3D, and TSM to PSL using the official PSL Dictionary augmented with spatial and photometric transformations. Their reported C3D accuracy of 93.33% represents the highest result in previous PSL video recognition work. However, as demonstrated in our experiments, this performance is heavily influenced by signer-inclusive random splitting on a dataset containing only two clips per class from a single signer. Under signer-disjoint evaluation, our own structurally similar model achieves only 12.88% accuracy. In later work, the same authors [14] transformed sign videos into static landmark trajectory images and reported 92.5% accuracy using an ensemble CNN approach. While effective numerically, this method removes temporal ordering information and excludes non-manual features such as facial expressions and body posture, both of which contribute linguistic meaning in PSL. To the best of our knowledge, our work is the first to evaluate dynamic PSL recognition using a strictly signer-disjoint protocol.

CNN and 3D-CNN Architectures for SLR.

Most isolated dynamic SLR systems follow a two-stage architecture in which CNNs extract spatial features from individual frames, while LSTM or BiLSTM networks model temporal relationships across sequences [11, 24]. Architectures such as C3D [21] and I3D [20] extend this approach by jointly learning spatial and temporal information through 3D convolutions. I3D, in particular, became widely adopted after Li et al. [31] showed that it outperformed graph-based methods for word-level ASL recognition.

The Temporal Shift Module (TSM) [22] offers a more computationally efficient alternative by shifting feature channels across adjacent frames to approximate temporal reasoning without full 3D convolutions. Despite these advances, such architectures still face a major limitation in low-resource settings. When training data is limited, models tend to overfit signer-specific appearance features instead of learning generalized gesture patterns. Marc et al. [2] demonstrated this effect on the LSA64 dataset, showing that signer-inclusive random splits can inflate reported accuracy by 30–50 percentage points compared to signer-independent evaluation. Our results on

PSL-104 confirm the same pattern.

Signer Independence and Evaluation Methodology.

The distinction between signer-dependent and signer-independent evaluation is one of the most important methodological considerations in SLR research, yet it is often neglected [9, 10]. Signer-dependent evaluation measures performance on signers already observed during training, allowing models to exploit identity-specific visual characteristics rather than genuine gesture representations. In contrast, signer-independent evaluation tests models on entirely unseen signers and therefore reflects realistic deployment conditions.

Previous studies have shown that the performance gap between these settings can be substantial [10]. Models reporting accuracies above 90% in signer-dependent settings may fall below 50% when evaluated signer-independently on small datasets. For datasets with limited numbers of signers, Leave-One-Signer-Out (LOSO) cross-validation is commonly recommended [2]. To date, no PSL-specific study has adopted a fully signer-disjoint evaluation protocol. Our work introduces this protocol for PSL-104 and applies it consistently across all experiments.

Transfer Learning and Large Pretrained Models for SLR.

The limitations of scratch-trained models in low-resource sign language settings have encouraged increased use of transfer learning. Selvaraj et al. [12] showed that pretrained models transferred from Indian Sign Language improve recognition accuracy across several low-resource sign languages by 2–18

More recently, visual-language pretraining methods based on CLIP [18] have shown strong potential for SLR tasks. CLIP was trained on 400 million image-text pairs and therefore learns highly general visual representations, including detailed representations of human hands under varied appearances, lighting conditions, and skin tones. These properties are especially valuable for sign language datasets, which are often small and visually limited. Zhou et al. [29] demonstrated that CLIP-inspired pretraining provides effective semantic representations for gloss-free sign language translation. Building on this idea, Luqman [15] proposed SignVLM, which combines a frozen CLIP ViT-L/14 encoder with a trainable EVL temporal decoder [19]. SignVLM achieved state-of-the-art performance across several sign language benchmarks, including KArSL, WLASL, LSA64, and AUTSL, particularly in low-data settings. Our work provides the first evaluation of SignVLM on PSL and demonstrates that CLIP-based transfer learning can substantially improve signer-independent PSL recognition.

Sentence-Level SLR and NLP Integration.

Although this study focuses on isolated word recognition, sentence-level sign language translation remains the broader long-term objective. Continuous SLR systems process uninterrupted signing streams and typically combine isolated recognition modules with temporal segmentation and language modelling [8]. Camgoz et al. [23] showed that transformer-based architectures can jointly learn recognition and translation, outperforming traditional pipeline approaches when sufficient training data is available.

For PSL, Khan et al. [17] proposed an NLP-based translation framework for converting English text into PSL glosses. This provides a practical downstream component that can be integrated with our isolated word recognition system. We therefore identify the combination of SignVLM-based recognition and NLP-driven translation as a feasible next step toward sentence-level PSL understanding.

Table 1 summarises the key studies discussed above, highlighting the input modality, dataset scope, and reported performance of each. A clear pattern emerges: prior PSL work either targets static images with small vocabularies, or applies video models without signer-disjoint evaluation. Our work addresses both gaps simultaneously.

Table 1: Summary of prior PSL and Urdu Sign Language recognition studies.

Study	**Input**	**Classes**	**Samples**	**SI?**	**Acc.**
Static Image-Based					
Halim & Abbas (2014)	Image	PSL signs	–	✗	91.0%
Kanwal et al. (2014)	Image	PSL signs	–	✗	90.0%
Nasir et al. (2014)	Image	PSL signs	–	✗	97/86%
Sami et al. (2014)	Image	37 alpha	–	✗	75.0%
Husnain et al. (2019)	Image	48 classes	38,400	✗	96/98%
Imran et al. (2021)	Image	37 alpha	1,480	✗	90.0%
Arooj et al. (2024)	Kinect	37 classes	–	✗	98.7%
Video-Based					
Hamza & Wali (2023)	Video	80 signs	160	✗	93.3%
Hamza & Wali (2025)	Traj.	100 signs	–	✗	92.5%
NLP / Translation					
Khan et al. (2020)	Text	Glosses	2k	✗	0.78†
This Work					
Ours (2026)	**Video**	**104 classes**	**1,872/6,974**	✓	**85.1%**

†BLEU score. SI? = Signer-Independent evaluation. alpha = alphabet. Traj. = Landmark trajectory.

3 Dataset: PSL-104

3.1 Motivation: The PSL Data Problem

The only existing publicly available PSL video resource is the official PSL Dictionary [16], which provides exactly two video samples per sign, both recorded by the same signer under identical conditions. This corpus has two structural deficiencies that make it unsuitable as a standalone training and evaluation resource: (1) the volume is too small to train any video recognition model, and (2) the absence of signer diversity means any model trained on it has no exposure to the inter-signer variation it will encounter in deployment.

Prior work [13, 14] uses this corpus with augmentation and random splitting, which inflates reported accuracy by allowing the model to exploit identity cues from the single training signer. Our work begins from a fundamentally different starting point: a purpose-built, multi-signer video corpus constructed to support genuine generalization.

3.2 Dataset Construction

3.2.1 Recording Protocol

The PSL-104 dataset was constructed by our team across two FYP cohorts. The recording protocol was standardized as follows:

- **Resolution and format:** All videos recorded at 1278×780 pixels, saved in .mp4 format.
- **Distance:** Signer positioned at a standardized shoulder distance of 2 metres from the camera.
- **Signers:** 6 signers total, each contributing 3 videos per class per signer.
- **Environmental diversity:** Each member records across varied environments (different rooms, outdoor settings) and varied lighting conditions (natural light, artificial light, mixed) to simulate real-world deployment diversity.
- **Signing variation:** Within each signer's 3 recordings per class, natural variation in signing speed, hand position, and articulation style is preserved rather than scripted.

This protocol yields 18 distinct original clips per class (6 signers $\times$ 3 videos), compared to the 2 clips per class (single signer) available from the PSL Dictionary. The 9$\times$ increase in per-class samples, combined with genuine signer diversity, makes signer-disjoint training and evaluation possible for the first time on PSL.

3.2.2 Class Coverage

PSL-104 covers 104 sign classes spanning English alphabet letters (A–Z, with hand orientation variants), Urdu/Arabic characters, and common vocabulary words. This breadth ensures the dataset covers the signs most frequently encountered in practical PSL communication.

3.3 Signer-Disjoint Splitting

A critical methodological contribution of this work is the enforcement of **signer-disjoint evaluation**. The six signers are assigned exclusively to one of three splits (train, validation, test), such that no signer's videos appear in more than one split. Formally:

$$\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{val}} = \emptyset, \quad \mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{test}} = \emptyset, \quad \mathcal{S}_{\text{val}} \cap \mathcal{S}_{\text{test}} = \emptyset \tag{1}$$

This mirrors the Leave-One-Signer-Out (LOSO) evaluation protocol recommended in the literature [2] and is the minimum requirement for results to have practical validity.

3.4 Video Augmentation Pipeline

With 15 original training clips per class before augmentation (after applying the signer-disjoint split), the training set is insufficient for deep learning models to achieve robust generalization. We developed a principled video augmentation pipeline (`augmentation_combined.py`) that applies stochastic spatial, photometric, and temporal transforms to generate synthetic variation while preserving sign semantics.

3.4.1 Augmentation Transforms

The pipeline applies 1–2 randomly selected transforms per video from the following set:

- **Lower-body crop** (`crop_ratio=0.2`): Removes the bottom 20% of each frame and resizes back to original resolution. Focuses the model on hands/torso and reduces irrelevant background variation.
- **Scale (zoom in/out)** (`scale_factor` $\in \{0.8, 1.2\}$): Resizes frames by the scale factor, then center-crops or zero-pads back to original size. Simulates camera zoom and subject distance variation.
- **Brightness** (`factor` $\sim$ Uniform(0.7, 1.4)): Global multiplicative intensity adjustment via `cv2.convertScaleAbs`. Simulates different lighting and exposure conditions.
- **Rotation** (`angle` $\sim$ Uniform($-10°$, $+10°$)): Rotates frames around image centre with reflection border mode. Simulates small camera tilt or signer lean.
- **Brightness-Saturation-Hue (BSH)**: Adjusts HSV channels; brightness 85–90%, saturation 110–115%, hue ± 10 degrees. Simulates colour/white-balance differences across cameras and skin tones.
- **Frame jitter** (`jitter_offset` $\sim$ Uniform(1, 3)): For each target frame index, applies a small random temporal offset. Simulates frame-rate variation and temporal misalignment.
- **Random frame drop** (`drop_rate` $\sim$ Uniform(0.05, 0.15)): Randomly removes up to 15% of frames, with a safety floor requiring at least 70% of frames to be retained. Simulates dropped frames from capture pipelines.

A minimum sequence length of 30 frames is enforced by repeating the final frame, ensuring all samples meet the minimum temporal length requirement.

3.4.2 Augmentation Application and Dataset Scale

Augmentation is applied to all three splits. For the training set, 30 augmented clips are generated per original clip (30 augmentations $\times$ 15 original = 450 augmented per class, for a total training set of 15 + 30 = 45 clips per class from the original videos). For validation and test sets, augmentation is applied to generate richer signer-disjoint evaluation pools, enabling statistically meaningful per-class accuracy estimates rather than single-sample test accuracy.

The resulting dataset statistics are summarized in Table 2.

Table 2: PSL-104 dataset statistics after augmentation.

Split	**Classes**	**Original Clips**	**After Augmentation**	**Avg./Class**
Training	104	1,560	4,680	45
Validation	104	312	1,248	12
Test	104	104	1,046	$\approx$10
Total	104	**1,976**	**6,974**	—

4 Methodology

4.1 Baseline: 3D-CNN + Residual Blocks + BiLSTM (Scratch)

4.1.1 Architecture

The baseline model is a custom temporal-spatial feature extraction network trained entirely from random initialization. It consists of three stages:

Stage 1 — Spatial Feature Extraction: Three Conv3D layers with batch normalization and ReLU activations progressively increase channel depth from 3 to 128. Two Residual Blocks (each with two $3 \times 3 \times 3$ convolutions and a skip connection) provide residual gradient flow. Adaptive average pooling projects spatial dimensions to 1×1, yielding a temporal sequence of 128-dimensional frame-level feature vectors.

Stage 2 — Temporal Modelling (BiLSTM): A 2-layer Bidirectional LSTM (hidden size 256 per direction) processes the feature sequence, producing a 512-dimensional summary at the final timestep.

Stage 3 — Classification: A linear projection maps the 512-dimensional representation to 104 class logits. Total trainable parameters: $\approx$2.5M.

4.1.2 Training Configuration

Table 3: 3D-CNN + BiLSTM training hyperparameters.

Hyperparameter	Value
Input resolution	112×112
Frames sampled	64
Batch size	8
Optimizer	AdamW
Learning rate	1e-4
LR schedule	Cosine decay + 3-epoch warmup
Label smoothing	0.04
Gradient clipping	1.0
Sampling strategy	WeightedRandomSampler
Early stopping patience	12 epochs
Mixed precision	AMP (float16)

4.2 SignVLM: CLIP-Based Visual-Temporal Transformer

4.2.1 Architecture Overview

SignVLM [15] is a two-component visual-temporal architecture that decouples spatial feature extraction from temporal modelling.

CLIP Visual Encoder (Frozen): Each of the T uniformly sampled frames is independently encoded by CLIP ViT-L/14 [18] — a Vision Transformer with 307M parameters pretrained on 400 million image-text pairs. The 14×14 patch grid preserves fine spatial detail at each frame. The [CLS] token produces a 1024-dimensional per-frame embedding. This encoder is kept **frozen** throughout training.

EVL Temporal Decoder (Trainable): The Efficient Video Learning (EVL) module [19] is a lightweight transformer decoder that models temporal dependencies across the sequence of CLIP embeddings using temporal convolutions, cross-frame attention, multi-head self-attention, and learnable classification tokens. Only the EVL decoder and the final linear classifier are updated during training.

4.2.2 PSL-Specific Adaptations

Several adaptations were required to deploy SignVLM on PSL-104:

- **Unicode path handling:** OpenCV's `VideoCapture()` silently fails on file paths containing Arabic/Urdu Unicode characters on Windows. We replaced the video loader with a PyAV (FFmpeg) implementation that handles Unicode correctly, resolving a silent data-loss failure for all Urdu-script class videos.
- **Offline frame pre-extraction:** All videos were pre-extracted to individual JPEG frames on disk, reducing data loading time by approximately $16\times$ and increasing GPU utilization from $\sim$30% to $\sim$85–90%.

- **CLIP normalization:** Mean = [0.4815, 0.4578, 0.4082], std = [0.2686, 0.2613, 0.2758] applied consistently across all clips.
- **Split file generation:** Dedicated scripts generate signer-disjoint split files in the `<path>\t<label>` format expected by SignVLM's dataloader.

4.2.3 Training Configuration

Table 4: SignVLM training hyperparameters.

Parameter	**Value**
Visual backbone	CLIP ViT-L/14 (frozen, 307M params)
Temporal decoder	EVL (trainable)
Spatial resolution	224×224
Frames per clip	16 or 24 (ablated)
Physical batch size	4
Gradient accumulation	4 steps (effective batch = 16)
Optimizer	AdamW
Peak learning rate	3e-5
LR schedule	Cosine decay
Training steps	10,000
Mixed precision	AMP (bfloat16)
GPU	NVIDIA RTX 3060 12GB

5 Experimental Results

5.1 Baseline: 3D-CNN + BiLSTM Failure Under Signer-Disjoint Evaluation

The 3D-CNN + BiLSTM baseline was trained for 30 epochs under signer-disjoint conditions. Table 5 summarises the training dynamics.

Table 5: 3D-CNN + BiLSTM training dynamics (signer-disjoint, scratch training).

Epoch	**Train Loss**	**Train Acc (%)**	**Val Loss**	**Val Acc (%)**
1	4.647	1.09	4.645	0.96
5	4.646	0.83	4.644	1.28
10	4.640	1.41	4.640	1.60
18	4.625	1.47	4.625	**12.88**
25	4.589	3.01	4.598	1.60
30	4.533	3.27	4.556	1.92

The model plateaued at 12.88% validation accuracy, barely above the 0.96% random-chance baseline for 104 classes. The near-flat loss ($\approx \log(104) \approx 4.644$ nats throughout) confirms that no meaningful learning occurred. This outcome is analysed theoretically in Section 6.

5.2 SignVLM: Main Results

Table 6 reports SignVLM's validation and test accuracy under the two frame sampling settings. All results are under signer-disjoint evaluation with the full augmented training set.

Table 6: SignVLM accuracy across frame sampling settings (full augmented training set, signer-disjoint).

Frames/Clip	Val Top-1 (%)	Val Top-5 (%)	Test Top-1 (%)
16	80.34	∼91	77.58
24	**83.49**	**∼93**	**83.12**

Increasing the temporal context from 16 to 24 frames per clip provides a consistent improvement of ∼3–6 percentage points across both splits, reflecting the benefit of denser temporal sampling for dynamic signs whose motion extends over longer durations.

5.3 N-Shot Ablation: Data Efficiency of SignVLM

To assess SignVLM's sample efficiency, we conduct an N-shot ablation study where only the first N distinct original videos per class are used for training (no augmented clips), with all runs using 24 frames per clip. The full-shot baseline additionally compares performance with and without augmentation.

Table 7: N-shot ablation for SignVLM (24 frames/clip, signer-disjoint). All N-shot runs use original videos only; "Full + Aug" adds 30 augmented clips per original.

Training Setting	Clips/Class	Val Top-1 (%)	Test Top-1 (%)
4-shot (original only)	4	81.25	80.13
8-shot (original only)	8	84.29	83.71
Full-shot (original only)	15	85.14	84.21
Full-shot + Augmentation	45	83.49	83.12
3D-CNN + BiLSTM (scratch)	15	12.88	—

Several observations stand out. First, SignVLM at 4 shots (81.25% val) already far exceeds the best scratch-trained baseline at full training (12.88%), confirming that CLIP's pretrained visual representations dramatically reduce the data requirements. Second, diminishing returns appear between 8-shot (84.29%) and full-shot (85.14%), suggesting the architecture converges to a near-optimal feature utilisation point with 8 original clips per class. Third, full-shot with augmentation (83.49%) slightly underperforms full-shot without augmentation (85.14%), which we attribute to the augmented clips introducing distribution shift relative to the validation and test signers; the original clips better represent the true signer distribution.

5.4 Comparative Summary

Table 8 consolidates the central result of this paper. The 29× accuracy collapse from signer-inclusive to signer-disjoint evaluation for the 3D-CNN baseline directly quantifies the generalization gap. SignVLM bridges this gap, achieving 83.49% under the same rigorous signer-disjoint conditions where the baseline achieves 12.88%.

Table 8: Main results: all models, evaluation protocols, and metrics on PSL-104.

Model	Eval Protocol	Data	Val Acc (%)	Test Acc (%)
3D-CNN+BiLSTM (Prior FYP)	Signer-inclusive	Original	83.89	—
3D-CNN+BiLSTM (Ours)	Signer-disjoint	Original	12.88	—
SignVLM 16-frame (Ours)	Signer-disjoint	Aug	80.34	77.58
SignVLM 24-frame (Ours)	Signer-disjoint	Aug	83.49	83.12
SignVLM 24-frame (Ours)	**Signer-disjoint**	**Original**	**85.14**	**84.21**
Random chance	—	—	0.96	—

6 Analysis

6.1 Why Scratch-Trained 3D-CNNs Fail

The near-chance accuracy (12.88%) of the 3D-CNN + BiLSTM under signer-disjoint evaluation is not a consequence of poor architecture design — the model is structurally sound and would perform well under signer-inclusive evaluation (as evidenced by the 83.89% figure from prior work using the same architecture class). The failure reflects three compounding statistical obstacles:

(1) Catastrophic data scarcity. Table 9 places PSL-104 training data in context. With ∼15 original clips per class, our training set is 40× smaller in samples-per-class than the minimum viable training condition for from-scratch 3D-CNN training (Kinetics-400 at ∼600 clips/class). No amount of architectural sophistication overcomes this fundamental statistical bottleneck.

Table 9: Training data comparison: PSL-104 vs. standard video recognition benchmarks.

Dataset	Total Clips	Classes	Clips/Class	Purpose
Sports-1M	1,000,000	487	∼2,050	C3D pretraining
Kinetics-400	240,000	400	∼600	I3D/SlowFast pretraining
WLASL	25,000	1,000	25	ASL recognition
LSA64	3,200	64	50	SLR
PSL-104 (Ours)	**1,560**	**104**	**∼15**	PSL recognition

(2) Inverted parameter-to-sample ratio. With ≈2.5M trainable parameters and 1,560 training samples, the parameter-to-sample ratio is approximately 1,603:1. Conventional deep learning practice requires at minimum ∼10 samples per parameter for gradient-based convergence; we have roughly the inverse.

(3) Signer-disjoint evaluation removes the shortcut. Under signer-inclusive splits (as in prior work), a model can achieve high test accuracy by memorising signer-specific appearance cues — background, skin tone, hand size, recording environment — without learning any generalisable sign patterns. The accuracy drop from 83.89% (signer-inclusive) to 12.88% (signer-disjoint) for the same architecture class quantifies exactly how much prior accuracy figures were inflated by this leakage. Marc et al. [2] report identical patterns on LSA64, confirming this as a universal failure mode of CNN-LSTM pipelines on small sign language datasets.

6.2 Why SignVLM Succeeds: Transfer from Scale

(1) CLIP's visual prior eliminates the data bootstrapping problem. CLIP ViT-L/14 was pretrained on 400 million image-text pairs. By the time it processes PSL videos, it already possesses rich spatial understanding of human hands and fingers across diverse appearances, lighting conditions, and backgrounds — all invariances that a scratch-trained model must learn from PSL data alone. The model does not need to learn what a hand looks like; it only needs to learn how PSL-specific hand motions map to class labels.

(2) The EVL decoder is parameter-efficient by design. Unlike a full video transformer that would require tens of thousands of training videos, EVL's lightweight temporal convolutions and cross-frame attention add a comparatively small number of trainable parameters on top of the frozen backbone. This efficient parameter allocation is precisely why 4 training clips per class is sufficient for 81.25% validation accuracy.

(3) CLIP representations are inherently signer-agnostic. CLIP's pretraining objective — aligning images with text descriptions — produces representations that abstract away identity-specific appearance. A description such as "a hand making a specific gesture" applies regardless of the identity of the signer. CLIP's features therefore encode a degree of signer invariance before any PSL-specific fine-tuning, which is precisely the property most needed for signer-disjoint generalisation.

(4) Cross-benchmark validation. As reported in the SignVLM paper [15], the architecture consistently outperforms state-of-the-art models across KArSL, WLASL, and LSA64, with the performance advantage being most pronounced in low-data settings. Our PSL results with ∼15 original clips/class extend this finding to a previously untested language.

7 Comparison to Prior PSL-Specific Work

Table 10 situates our results within the landscape of prior PSL recognition work. The comparison requires careful interpretation of evaluation conditions because, as we demonstrate, reported accuracy figures are only meaningful when stated alongside the evaluation protocol.

Table 10: Comparison to prior PSL recognition methods. SD = Signer-Disjoint evaluation. SI = Signer-Inclusive. "Static" = operates on still images only.

Method	Year	Input	Classes	Eval	Acc (%)
Imran et al. [3]	2021	Static image	36	SI	∼85
Arooj et al. [4]	2024	Static image	36	SI	∼88
Hamza & Wali [13]	2023	Video (C3D)	100	SI	93.33
Hamza & Wali [14]	2025	Video→Image	100	SI	92.5
3D-CNN+BiLSTM (Ours)	2026	Video	104	**SD**	12.88
SignVLM (Ours)	**2026**	**Video**	**104**	**SD**	**85.14**

7.1 Against Static Alphabet Methods (Imran 2021; Arooj 2024)

Imran et al. [3] presented a dataset of PSL hand configurations for Urdu alphabet characters and evaluated machine learning classifiers on static images. **Arooj et al.** [4] extended this line of work using a CNN-SIFT hybrid framework with Kinect sensor data, targeting the same static Urdu alphabet. Both works are limited to static alphabets in three critical ways that our approach overcomes:

- **Temporal information.** Static recognition cannot handle dynamic signs where meaning depends on the motion trajectory, not just the hand shape at a single moment. PSL, like all natural sign languages, is predominantly dynamic. Our approach processes the full video sequence, with the EVL temporal decoder explicitly modelling motion dependencies across 24 frames.
- **Vocabulary coverage.** Both works cover only Urdu alphabet letters (36 classes at most). Our system covers 104 classes including full vocabulary words, making it substantially more useful for real communication support.
- **Evaluation validity.** Neither work addresses signer-independent evaluation. A static image classifier that learns the appearance of hands from one or two signers cannot be assumed to generalise to new signers. Our signer-disjoint protocol provides the first rigorous assessment of generalization for PSL.

7.2 Against Video-Based PSL Methods (Hamza & Wali 2023, 2025)

Hamza & Wali (2023) [13] applied C3D, I3D, and TSM to PSL video. While this is the closest prior work in modality and scope, several fundamental limitations distinguish it from our approach:

Evaluation inflation. The 93.33% reported for C3D uses signer-inclusive random splitting on a corpus with 2 clips/class from a single signer. Our signer-disjoint experiment with a structurally similar 3D-CNN achieves 12.88%, quantifying a 90-point inflation attributable entirely to signer leakage. The comparison is stark: the same class of model, the same task, a difference of 90 percentage points explained solely by evaluation methodology.

Model-specific weaknesses.

- *C3D* was pretrained on Sports-1M in 2014, an action recognition corpus of coarse body-motion categories (sports). Its visual representations are not optimised for fine-grained hand pose discrimination, which is the primary discriminative signal in sign language. SignVLM's CLIP backbone was pretrained on 400M diverse image-text pairs including abundant hand imagery.
- *I3D* requires full fine-tuning on PSL data. With 2 original clips/class, this fine-tuning is essentially memorisation of the training signer's appearance rather than learning of sign-specific patterns.
- *TSM* approximates temporal modelling by shifting feature channels between adjacent frames. This provides only local temporal context and cannot model dependencies between frames

far apart in the sequence — for example, between a sign's initial handshape and its final position. SignVLM's cross-frame attention explicitly attends over all frames simultaneously.

Hamza & Wali (2025) [14] converted videos to static landmark trajectory images, achieving 92.5% with ensemble CNNs. Beyond the same evaluation inflation issue, this approach has two structural limitations:

- **Temporal ordering is destroyed.** A trajectory image shows where keypoints moved but not the order in which they moved. Any sign pair that shares a trajectory but differs in temporal order (e.g., a sign that opens then closes vs. one that closes then opens) is indistinguishable to this representation.
- **Non-manual cues are completely discarded.** The 21 MediaPipe hand keypoints used for trajectory construction contain no information about facial expression, head movement, or upper body posture — all of which carry grammatical meaning in natural PSL (e.g., raised eyebrows for yes/no questions, negative head shake, intensity modifiers). SignVLM processes raw RGB frames and the CLIP encoder retains these cues in its representations.

8 Discussion and Limitations

8.1 Practical Significance of Results

SignVLM achieves 83.49% validation and 83.12% test accuracy on a 104-class PSL dataset under strictly signer-disjoint evaluation conditions, demonstrating for the first time that near-deployment-grade PSL recognition is achievable without signer-specific training data. The N-shot results are particularly significant for practical deployment: 81.25% accuracy at 4 shots per class means the system can be adapted to new signing contexts with minimal data collection, lowering the barrier to extending the vocabulary or deploying to new domains.

8.2 Limitations

Vocabulary scope. 104 classes covers a representative but limited subset of PSL vocabulary. Expanding to hundreds or thousands of signs will require proportionally more data collection and will test the architecture's scalability beyond the current low-resource regime.

Augmentation vs. original data gap. The slight underperformance of the augmented training setting (83.49%) relative to original-only (85.14%) suggests the current augmentation pipeline introduces some distribution shift. Future work should explore augmentations more closely calibrated to the signer-to-signer variation actually observed in the dataset (e.g., signer-swap augmentation using pose retargeting).

Sentence-level recognition not yet addressed. The current system operates in an isolated word recognition setting. Real-world PSL communication is continuous and sentence-level, requiring segmentation of the continuous signing stream into individual signs and then composition of recognized signs into grammatically coherent text.

8.3 Path to Sentence-Level PSL Translation

The most impactful near-term extension is moving from isolated word recognition to sentence-level PSL translation. We propose a concrete pipeline:

1. **Sliding window sign segmentation:** A sliding window over the webcam video stream activates SignVLM inference every Δt seconds. High-confidence individual sign predictions are accumulated into a sign sequence buffer.
2. **Gloss sequence:** The recognized sign labels form a PSL gloss sequence (an ordered list of sign names).
3. **NLP-based translation:** The gloss sequence is passed to an NLP translation module. Khan et al. [17] proposed a machine translation model specifically for English-to-PSL and PSL-gloss-to-English translation; integrating their work with our recognition output provides a complete sign-to-sentence pipeline in both Urdu and English.

Future dataset expansion. Scaling this pipeline to practical PSL communication requires a substantially larger vocabulary. We plan to expand the PSL-104 corpus by adding new sign classes while maintaining the 6-signer diversity protocol, targeting 300–500 classes in the next collection phase. Maintaining signer diversity during expansion ensures the generalization gains demonstrated in this paper are preserved at larger scale.

9 Conclusion

This paper presented a systematic investigation of the generalization gap in Pakistan Sign Language recognition and a practical solution to it. We made two core contributions: a purpose-built multi-signer PSL-104 video dataset that, for the first time, supports rigorous signer-disjoint evaluation of dynamic PSL recognition, and an adaptation of the SignVLM architecture that exploits CLIP's large-scale pretrained visual representations to achieve strong performance in this low-resource setting.

Our central empirical finding is that the generalization gap in PSL recognition is substantially larger than prior work has acknowledged: a 3D-CNN + BiLSTM model achieves 83.89% under the signer-inclusive evaluation prevalent in the literature but collapses to 12.88% under signer-disjoint evaluation — a 29× drop explained entirely by signer-identity leakage in the evaluation protocol. SignVLM recovers this gap, achieving 85.14% top-1 test accuracy under signer-disjoint conditions, and maintains 81.25% accuracy even with only 4 training clips per class.

We further showed that all existing PSL-specific methods — static alphabet classifiers [3, 4], C3D/I3D/TSM video models [13], and video-to-image conversion pipelines [14] — share the same evaluation flaw and additionally either lack temporal modelling or discard non-manual linguistic cues. SignVLM addresses all of these limitations simultaneously: it processes full video sequences via transformer-based temporal attention, preserves non-manual cues through raw RGB input to a rich semantic encoder, and requires no task-specific pretraining, relying instead on CLIP's broadly pretrained visual representations.

The results confirm that for low-resource sign languages with fewer than 50 training clips per class, pretrained large-scale visual representations are not merely beneficial but essential: from-scratch training is not a viable path regardless of architectural sophistication. The PSL recognition problem requires the leverage of models trained at internet scale before task-specific fine-tuning can succeed.

The path forward is clear: expand the PSL-104 corpus with additional signers and classes, integrate SignVLM's word-level recognition with NLP-based sentence generation [17], and deploy a real-time PSL-to-text pipeline that serves Pakistan's one-million-strong deaf community.

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